

Theory and implementation of reusable real-time TDDFT in Aion

Bare length and velocity gauges, covariant P0/P0+E1 dynamics,
currents, energies, and connection-aware propagation

Aion project

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Abstract

This note fixes the physical definitions, sign and orientation conventions, equations of motion, observables, numerical algorithms, and software boundaries for the proposed Aion 0.2 real-time time-dependent density-functional theory (RT-TDDFT) implementation. It treats four formulations on one common finite-basis electronic reference: bare length gauge, bare velocity gauge, covariant P0, and covariant P0+E1. Particular attention is paid to quantities that cannot be transferred uncritically between formulations: the primary current, the mechanical-current diagnostic, matter and generator energies, source work, and the evolving-metric connection. The production propagation algorithm is self-consistent exponential midpoint (SCEM), with a connection-aware transport for the covariant formulations and no time-dependent Löwdin frame. The final sections map these definitions to a reusable API, storage schema, and validation hierarchy. This is a normative design document; it does not claim that the refactor described here has already been implemented.

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1 Document status and scope

1.1 Meaning of status labels

Four kinds of statements are distinguished throughout this note.

Established theory

Standard RT-TDDFT, minimal coupling, gauge transformation, evolving-Hilbert- space geometry, and rational approximation results supported by the cited literature.

Normative Aion 0.2

An agreed definition or software/numerical contract that the refactor must implement.

Draft-validated Physics or algorithms exercised in the present research tree, but not yet available through the clean 0.2 API or fully covered by its tests.

Planned/unvalidated

Agreed work whose complete reusable implementation and validation remain to be done.

Status — Normative Aion 0.2. When this note uses *must* or gives an unqualified stored name, it specifies the intended 0.2 contract. The implementation plan in `docs/refactor_implementation_plan.md` controls the order and acceptance gates. That plan is deliberately mutable; changes to the physics conventions require an explicit reviewed decision.

1.2 Validated physical domain

The first release targets finite, fixed-nucleus, all-electron molecules in restricted spin-summed Kohn–Sham DFT, with pure LDA or GGA functionals and prescribed spatially uniform electric fields. Atomic units are used internally and the numerical precision is double real/complex precision on CPU and GPU. The nuclear attraction is local.

The following are explicitly outside the first physics stack:

- pseudopotentials, effective-core potentials, and nonlocal ionic operators;
- hybrid and exact-exchange current corrections;
- unrestricted, noncollinear, or relativistic spin dynamics;
- moving nuclei and their basis-velocity connection;
- periodic boundary conditions;
- Maxwell backreaction and adaptive time grids.

These cases must be rejected rather than silently treated with formulas valid only for local all-electron Hamiltonians. The source and observable interfaces are nevertheless designed so that spatially varying fields, aloof relativistic projectiles, EELS, and possibly periodic velocity gauge can be added later without redefining the present uniform-field theory. They are extension seams, not present work packages.

2 Notation and conventions

2.1 Atomic basis and density

Let $\{|\phi_\mu(t)\rangle\}$ be a generally time-dependent nonorthogonal atomic-orbital (AO) basis. The overlap and occupied orbitals are

$$S_{\mu\nu}(t) = \langle\phi_\mu(t)|\phi_\nu(t)\rangle, \quad |\psi_i(t)\rangle = \sum_\mu |\phi_\mu(t)\rangle C_{\mu i}(t). \quad (1)$$

With occupations collected in f , the canonical AO density convention is

$$P = C f C^\dagger, \quad N = \text{Re Tr}(PS), \quad \langle O \rangle = \text{Re Tr}(PO). \quad (2)$$

The occupations contain the complete spin multiplicity. No contraction may insert an implicit closed-shell factor of two. A different covariant/dual or orthonormal representation is a different type and requires an explicit conversion.

Electronic charge and mass are kept symbolically as q and m ; electrons have $q = -1$, $m = 1$ in atomic units. We retain $\hbar$ in derivations to expose phases and signs, although $\hbar = 1$ in the implementation. The reduced vector potential is

$$\mathbf{a} = \frac{\mathbf{A}}{c}, \quad (3)$$

and the full stored name is `vector_potential_reduced`. This avoids an otherwise persistent ambiguity about factors of c .

2.2 Coordinates, origin, and molecular observables

Input coordinates are preserved exactly: Aion must never silently recenter a molecule. An explicit electromagnetic origin $\mathbf{O}$, defaulting to the input-coordinate origin, is stored. Electronic, fixed-nuclear, and total dipoles use this same origin:

$$\boldsymbol{\mu}_e = q \text{Re Tr}[P(\mathbf{r} - \mathbf{O}S)], \quad (4)$$

$$\boldsymbol{\mu}_N = \sum_A Z_A (\mathbf{R}_A - \mathbf{O}), \quad (5)$$

$$\boldsymbol{\mu}_{\text{tot}} = \boldsymbol{\mu}_e + \boldsymbol{\mu}_N. \quad (6)$$

For the finite molecule, $\mathbf{J}$ denotes total charge current, not a volume-normalized current density.

2.3 Time grid and sampling location

For a fixed step, integer index is authoritative:

$$t_n = t_0 + n\Delta t, \quad n = 0, \dots, N. \quad (7)$$

There are N propagation intervals and $N + 1$ endpoint states. Floating-point time is stored as a convenience. State, dipole, current, and sampled energy records are endpoint-centred. Work increments, point power, and Ward residuals associated with a propagation interval are midpoint- or interval-centred and carry their own step/time arrays.

3 Electronic structure and the AO equation of motion

3.1 Field-free Kohn–Sham model

The all-electron molecular reference is a converged restricted Kohn–Sham calculation. For a pure LDA/GGA functional its field-free Fock/Kohn–Sham matrix is written

$$H_0[P] = T + V_{\text{nuc}} + J[P] + V_{\text{xc}}[P]. \quad (8)$$

The corresponding field-free energy decomposition is

$$T_s[P] = \text{Re Tr}(PT), \quad E_{\text{eN}}[P] = \text{Re Tr}(PV_{\text{nuc}}), \quad (9)$$

$$E_{\text{H}}[P] = \frac{1}{2} \text{Re Tr}(PJ[P]), \quad E_{\text{xc}}[P] = E_{\text{xc}}[n_P], \quad (10)$$

$$E_{\text{NN}} = \sum_{A < B} \frac{Z_A Z_B}{R_{AB}}, \quad U_0[P] = T_s + E_{\text{eN}} + E_{\text{H}} + E_{\text{xc}} + E_{\text{NN}}. \quad (11)$$

Real-time propagation is used within the Runge–Gross density–potential framework[1]; PBE is a representative pure GGA in the validated functional class[5]. The implementation uses PySCF for the RKS reference, numerical grid, effective potential, and functional energy[6]. The first release stores combined exchange–correlation energy; an exchange/correlation split is not a required observable.

3.2 Connection in a time-dependent nonorthogonal basis

Projection of the time-dependent Kohn–Sham equation into an evolving basis gives

$$i\hbar \left(S\dot{C} + \omega C \right) = H_{\text{eom}} C, \quad \omega_{\mu\nu} = \langle \phi_\mu | \dot{\phi}_\nu \rangle. \quad (12)$$

The geometry is not an optional correction. It imposes metric compatibility

$$\dot{S} = \omega + \omega^\dagger, \quad (13)$$

which guarantees preservation of the physical overlap norm under the exact equation. Artacho and O’Regan discuss why an evolving Hilbert space is geometrically different from a standard fixed-coordinate matrix ODE[4].

For implementation, define

$$\mathcal{A} = S^{-1} \left[-\omega - \frac{i}{\hbar} H_{\text{eom}} \right], \quad \dot{C} = \mathcal{A}C, \quad \dot{P} = \mathcal{A}P + P\mathcal{A}^\dagger. \quad (14)$$

Knowledge of $S(t)$ and $\omega(t)$ at arbitrary times removes any need to integrate a separate metric equation. It does not remove the need to transport the electronic state consistently between two different metrics, nor the need to solve the nonlinear midpoint Hamiltonian self-consistently.

Status — Normative Aion 0.2. Every formulation returns the explicit triple $(S, H_{\text{eom}}, \omega)$. The older implicit split between an intrinsic Hamiltonian and a residual connection is not a public contract.

4 Electromagnetic sources and gauge data

4.1 Potential-first representation

The minimal-coupling one-particle operator is

$$h(t) = \frac{1}{2m} (\mathbf{p} - q\mathbf{a}(\mathbf{r}, t))^2 + q\phi(\mathbf{r}, t) + v_0, \quad (15)$$

where $\mathbf{p} = -i\hbar\nabla$. The electric field is

$$\mathbf{E} = -\partial_t \mathbf{a} - \nabla \phi. \quad (16)$$

A gauge function χ acts as

$$\mathbf{a}' = \mathbf{a} + \nabla \chi, \quad \phi' = \phi - \dot{\chi}, \quad \psi' = \exp(iq\chi/\hbar)\psi. \quad (17)$$

The reusable source layer describes the physical field separately from a chosen gauge representation. It is potential-first and evaluable at arbitrary times, including every SCEM midpoint. A field-first convenience input is compiled to consistent potentials before propagation; $\mathbf{a}$ is never constructed by incremental numerical integration inside the electronic hot loop.

4.2 Projected node/link sample

For the covariant formulation, associate AO μ with an atomic anchor $a(\mu)$. The compact projected source sample is

$$\{\phi_a(t), L_{ab}(t), \dot{L}_{ab}(t)\}, \quad L_{ab} = \int_{\mathbf{R}_b}^{\mathbf{R}_a} \mathbf{a} \cdot d\boldsymbol{\ell}, \quad a < b. \quad (18)$$

The pair electromotive potential is

$$V_{ab} = -\dot{L}_{ab} - \phi_a + \phi_b. \quad (19)$$

Links and EMFs are oriented $b \rightarrow a$, while $I_{ab} > 0$ denotes charge flow $a \rightarrow b$. With the corresponding signed incidence matrix,

$$\dot{Q}_a + \sum_b I_{ab} = 0, \quad (20)$$

$$\mathbf{J}_{\text{graph}} = \sum_{a < b} I_{ab} (\mathbf{R}_b - \mathbf{R}_a), \quad (21)$$

$$\mathcal{P}_{\text{source}} = - \sum_{a < b} I_{ab} V_{ab}. \quad (22)$$

Under Eq. (17), $L'_{ab} = L_{ab} + \chi_a - \chi_b$, so Eq. (19) is invariant.

Multiple physical providers are added at the primitive potential/link level before projection or Wilson dressing. A temporal gate $g(t)$ acts on potentials,

$$\phi^g = g\phi, \quad L^g = gL, \quad \dot{L}^g = \dot{g}L + g\dot{L}. \quad (23)$$

Omitting the product-rule term would omit a real switching field.

4.3 Finite pulses and discrete kicks

A compactly supported resonant pulse may be defined in the vector potential:

$$\mathbf{a}(t) = a_0 \sin^2\left(\frac{\pi t}{T}\right) \sin(\Omega t + \varphi) \hat{\mathbf{e}}, \quad 0 \leq t \leq T, \quad (24)$$

with T an exact integer number of carrier cycles and $\mathbf{E} = -\dot{\mathbf{a}}$. The amplitude is normalized to the requested peak electric field. Equation (24) has $\mathbf{a}(0) = \mathbf{a}(T) = 0$ and zero net impulse. This is a property of this pulse family, not a restriction on the physical use of pulses.

If a maximum step is requested, choose

$$N = \lceil T/\Delta t_{\text{max}} \rceil, \quad \Delta t = T/N \quad (25)$$

so the boundary is exactly a grid point. General pulses may carry nonzero impulse. A simulation or analysis workflow may request a zero-impulse constraint when its scientific purpose requires one; the pulse abstraction itself imposes no such policy.

A delta kick is a uniquely identified discrete event, not a narrow sampled pulse. Events lie exactly on grid boundaries and are stored with pre- and post-event records at the same physical time, distinguished by side and sequence. The exact map is formulation-owned:

- bare LG applies the coefficient-space representation of $\exp(i\boldsymbol{\kappa} \cdot \mathbf{r})$;
- bare VG leaves coefficients continuous and makes the discontinuous change $\mathbf{a}(0^+) = -\boldsymbol{\kappa}$ in the adopted electronic sign convention;
- P0+E1 applies its covariant gauge/state event map.

Applied event identifiers are checkpointed, making a restarted event idempotent.

5 Bare length-gauge formulation

5.1 Equation of motion

For a uniform field in length gauge, $\mathbf{a} = 0$ and $\phi = -\mathbf{E} \cdot (\mathbf{r} - \mathbf{O})$. In a fixed AO basis,

$$S = S_0, \quad \omega = 0, \quad H_{\text{LG}}[P, t] = H_0[P] - q\mathbf{E}(t) \cdot (\mathbf{r} - \mathbf{O}S). \quad (26)$$

5.2 Dipole and currents

The primary current is the analytic time derivative of the length-gauge dipole:

$$\mu_{\text{e},\alpha} = q \operatorname{Re} \operatorname{Tr}[PR_\alpha], \quad R_\alpha = r_\alpha - O_\alpha S, \quad (27)$$

$$\begin{aligned} J_{\text{LG},\alpha} &= q \operatorname{Re} \operatorname{Tr}[\dot{P}R_\alpha] \\ &= \Re \left\{ \frac{iq}{\hbar} \operatorname{Tr} \left[P \left(H_{\text{LG}} S^{-1} R_\alpha - R_\alpha S^{-1} H_{\text{LG}} \right) \right] \right\}. \end{aligned} \quad (28)$$

The second line follows from Eq. (14) and is not a finite difference.

For comparison, a local all-electron ambient mechanical diagnostic is

$$J_{\text{para},\alpha} = \frac{q}{m} \operatorname{Re} \operatorname{Tr}(Pp_\alpha), \quad J_{\text{dia},\alpha} = 0, \quad \mathbf{J}_{\text{mech}} = \mathbf{J}_{\text{para}}. \quad (29)$$

The analytic dipole current is the primary LG observable. In a finite basis, Eq. (28) and Eq. (29) need not coincide; their difference measures finite-subspace failure of continuum commutator identities.

5.3 Energy

The length-gauge matter energy excludes the source coupling:

$$U_{\text{matter}}^{\text{LG}} = U_0[P]. \quad (30)$$

The electronic and fixed-nuclear scalar couplings and generator energy are

$$U_{\text{EM,e}}^{\text{LG}} = -\mathbf{E} \cdot \boldsymbol{\mu}_{\text{e}}, \quad U_{\text{EM,N}}^{\text{LG}} = -\mathbf{E} \cdot \boldsymbol{\mu}_{\text{N}}, \quad (31)$$

$$E_{\text{generator}}^{\text{LG}} = U_{\text{matter}}^{\text{LG}} + U_{\text{EM,e}}^{\text{LG}} + U_{\text{EM,N}}^{\text{LG}}. \quad (32)$$

For fixed nuclei,

$$\frac{dU_{\text{matter}}^{\text{LG}}}{dt} = \mathbf{E} \cdot \mathbf{J}_{\text{LG}}, \quad (33)$$

$$\frac{dE_{\text{generator}}^{\text{LG}}}{dt} = -\dot{\mathbf{E}} \cdot \boldsymbol{\mu}_{\text{tot}}. \quad (34)$$

These are different valid energies. The phrase *total energy* is therefore insufficient without the qualifier **matter** or **generator**. When the field vanishes at both selected endpoints, the final deposited matter energy and final generator-energy gain agree.

6 Bare velocity-gauge formulation

6.1 Hamiltonian

For $\phi = 0$ and spatially uniform $\mathbf{a}(t)$, expansion of Eq. (15) gives the AO matrix

$$H_{\text{VG}}[P, t] = H_0[P] - \frac{q}{m} \mathbf{a}(t) \cdot \mathbf{p} + \frac{q^2}{2m} |\mathbf{a}(t)|^2 S. \quad (35)$$

This is the local all-electron velocity-gauge form used in the NAO RT-TDDFT treatment of Pemmaraju *et al.*[3]. For a nonlocal ionic potential, an additional gauge-transformed nonlocal operator and corresponding current correction are required; they are outside scope.

With PySCF derivative overlaps, the implementation fixes and tests the canonical-momentum convention

$$p_\alpha = i\hbar\langle\nabla_\alpha\phi_\mu|\phi_\nu\rangle, \quad (36)$$

including its Hermiticity and sign.

6.2 Mechanical current

The bare VG primary current is the mechanical current,

$$J_{\text{para},\alpha} = \frac{q}{m} \text{Re Tr}(Pp_\alpha), \quad (37)$$

$$J_{\text{dia},\alpha} = -\frac{q^2}{m}Na_\alpha, \quad (38)$$

$$J_{\text{mech},\alpha} = J_{\text{para},\alpha} + J_{\text{dia},\alpha}. \quad (39)$$

The two pieces must be available separately. Plotting only $q \text{Re Tr}(Pp_\alpha)$ as the VG physical current omits the diamagnetic piece.

6.3 Mechanical kinetic and matter energy

The mechanical kinetic energy is

$$T_{\text{mech}}[P, t] = T_{\text{can}}[P] + T_{\text{linear}}[P, t] + T_{\text{dia}}[P, t], \quad (40)$$

$$T_{\text{can}} = \text{Re Tr}(PT),$$

$$T_{\text{linear}} = -\frac{q}{m}\mathbf{a} \cdot \text{Re Tr}(P\mathbf{p}),$$

$$T_{\text{dia}} = \frac{q^2}{2m}N|\mathbf{a}|^2.$$

The VG matter energy is

$$U_{\text{matter}}^{\text{VG}} = T_{\text{mech}} + E_{\text{eN}} + E_{\text{H}} + E_{\text{xc}} + E_{\text{NN}}. \quad (41)$$

In the pure VG representation it is also the electronic generator energy for the prescribed field, and

$$\frac{dU_{\text{matter}}^{\text{VG}}}{dt} = \mathbf{E} \cdot \mathbf{J}_{\text{mech}}. \quad (42)$$

An energy that retains only T_{can} contains a large, apparently anomalous oscillatory error during the pulse. Both the linear $\mathbf{a} \cdot \mathbf{p}$ and quadratic a^2 contributions in Eq. (40) are compulsory.

7 Covariant P0 formulation

7.1 Wilson-dressed overlap

Let each AO carry an anchor position $\mathbf{R}_{a(\mu)}$. With the link oriented from the anchor of ν to that of μ , define

$$\Theta_{\mu\nu}(t) = \exp\left[\frac{iq}{\hbar}L_{a(\mu)a(\nu)}(t)\right], \quad S_{\mu\nu}(t) = \Theta_{\mu\nu}(t)S_{\mu\nu}^0. \quad (43)$$

The inverse-dressed density passed to the field-free electronic-structure model is

$$P^0 = \Theta^* \odot P, \quad (44)$$

where $\odot$ is elementwise multiplication. If $H_0[P^0]$ is constructed by PySCF, the P0 equation-of-motion matrix is obtained by the forward dressing

$$H_{P0}[P, t] = \Theta \odot H_0[P^0]. \quad (45)$$

7.2 Projected connection

Introduce the diagonal site scalar-potential connection

$$\sigma_{\mu\nu} = \delta_{\mu\nu} \frac{iq}{\hbar} \phi_{a(\mu)}. \quad (46)$$

The site-covariant overlap derivative and P0 connection are

$$D_{\text{site}}S = \dot{S} + \sigma S - S\sigma, \quad (47)$$

$$\omega_{P0} = S\sigma + \frac{1}{2}D_{\text{site}}S. \quad (48)$$

Because $\sigma^\dagger = -\sigma$ for a real scalar potential, Eq. (48) satisfies $\omega_{P0} + \omega_{P0}^\dagger = \dot{S}$ exactly.

The P0 equation is consequently

$$i\hbar(S\dot{C} + \omega_{P0}C) = H_{P0}[P, t]C. \quad (49)$$

The scalar-potential and link time dependence is geometrical: it is carried by the projected connection and metric, not added again to Eq. (45).

7.3 Site charge and graph current

Let M_a be the fixed AO selector/projector associated with anchor a . The symmetrized source charge is

$$Q_a = \frac{q}{2} \text{Re Tr}[P\{M_a, S\}]. \quad (50)$$

For each stored pair, let S_{ab} denote its oriented overlap contribution and $H_{\text{source},ba}$ the corresponding explicit source block. Define

$$K_{\text{dyn}} = H_{\text{dyn}} - \frac{i\hbar}{2}D_{\text{site}}S. \quad (51)$$

Here H_{dyn} is the Hermitian matrix in the equivalent coefficient-velocity equation after the non-metric connection terms have been moved to its right-hand side. Thus $H_{\text{dyn}} = H_{P0}$ for P0 and $H_{\text{dyn}} = H_{P0} + V_{E1}$ for P0+E1. This algebraic definition does not alter the normative connection split in Eq. (57). The variational graph-current expression used by the covariant model is

$$\begin{aligned} I_{ab} = & \frac{q}{\hbar} \text{Im Tr} \left[P \left(S_{ab} S^{-1} K_{\text{dyn}} + K_{\text{dyn}}^\dagger S^{-1} S_{ab} \right) \right] \\ & + \frac{2q}{\hbar} \text{Im Tr}(P H_{\text{source},ba}). \end{aligned} \quad (52)$$

Together with Eqs. (20)–(22), this gives the P0 charge, current vector, and source power. The pair orientation and the definition of the source block are part of the data model, not choices made by individual analysis scripts.

8 P0+E1 correction

8.1 Central dipoles and variational connection

For the AO pair (μ, ν) , define the anchor-pair centre

$$\mathbf{R}_{\mu\nu} = \frac{1}{2} \left(\mathbf{R}_{a(\mu)} + \mathbf{R}_{a(\nu)} \right) \quad (53)$$

and field-free central dipole matrices

$$d_{\mu\nu}^\alpha = q \left[r_{\mu\nu}^\alpha - R_{\mu\nu}^\alpha S_{\mu\nu}^0 \right]. \quad (54)$$

Their dressed Hermitian form and E1 variational potential are

$$D_\alpha(t) = \text{Herm}[\Theta(t) \odot d^\alpha], \quad \text{Herm } X = \frac{1}{2}(X + X^\dagger), \quad (55)$$

$$V_{\text{E1}}[P, t] = - \sum_\alpha E_\alpha(t) D_\alpha(t). \quad (56)$$

The defining physics split is

$$\boxed{H_{\text{eom}} = H_{\text{P0}}, \quad \omega_{\text{P0+E1}} = \omega_{\text{P0}} + \frac{i}{\hbar} V_{\text{E1}}.} \quad (57)$$

Putting V_{E1} in both H_{eom} and ω double counts it. Equation (57) expresses the projection of the ambient electromagnetic connection into the P0+E1 equivariant subspace.

8.2 Covariant dipole

Writing $N_a = Q_a/q$, the electronic covariant dipole is

$$\mu_{\text{P0+E1}, \alpha} = q \sum_a N_a R_{a\alpha} + \text{Re Tr}(PD_\alpha). \quad (58)$$

The first term transports site charge between anchors; the second restores intra-/inter-site polarization retained at E1.

8.3 E1 residual and total source current

The reduced-vector-potential derivative of the dressed central dipole is

$$D_{\alpha; \beta} = \frac{\partial D_\alpha}{\partial a_\beta} = \text{Herm} \left[\frac{iq}{\hbar} (R_{a(\mu), \beta} - R_{a(\nu), \beta}) \Theta_{\mu\nu} d_{\mu\nu}^\alpha \right]. \quad (59)$$

The E1 residual source current is

$$J_{\text{E1}, \beta} = \frac{d}{dt} \text{Re Tr}(PD_\beta) + \sum_\alpha E_\alpha \text{Re Tr}(PD_{\alpha; \beta}). \quad (60)$$

Its first term is the intrinsic polarization rate and its second term is the phase response. Both pieces are recorded.

The total P0+E1 variational source current is

$$\mathbf{J}_{\text{src}} = \sum_{a < b} I_{ab}^{(\text{P0|E1})} (\mathbf{R}_b - \mathbf{R}_a) + \mathbf{J}_{\text{E1}}. \quad (61)$$

In Eq. (52), the coefficient/dynamical part sees the full P0+E1 equation, while the explicit source-Hamiltonian block is the base P0 source contribution. This separation prevents E1 double counting.

The defining checks are

$$\dot{Q}_a + \sum_b I_{ab} = 0, \quad (62)$$

$$\mathbf{J}_{\text{src}} = \frac{d\mu_{\text{P0+E1}}}{dt}, \quad (63)$$

$$\frac{dU_{\text{cov}}}{dt} = \mathbf{E} \cdot \mathbf{J}_{\text{src}}. \quad (64)$$

Status — Draft-validated. The current research tree contains the P0/P0+E1 tensors, connection split, site charges, pair-current expression, and analytic E1-current pieces. Their clean typed integration with the complete energy ledger and storage schema is planned, not yet complete.

9 Mechanical-current diagnostics

9.1 Generic local contraction

For an arbitrary density and prescribed local vector potential, define AO matrices

$$A_{\alpha,\mu\nu}^{\text{op}}(t) = \langle \phi_\mu | a_\alpha(\mathbf{r}, t) | \phi_\nu \rangle. \quad (65)$$

The reusable low-level diagnostic is

$$J_{\text{para},\alpha} = \frac{q}{m} \text{Re Tr}(P p_\alpha), \quad (66)$$

$$J_{\text{dia},\alpha} = -\frac{q^2}{m} \text{Re Tr}(P A_\alpha^{\text{op}}), \quad (67)$$

$$\mathbf{J}_{\text{mech}} = \mathbf{J}_{\text{para}} + \mathbf{J}_{\text{dia}}. \quad (68)$$

For a uniform field, $A_\alpha^{\text{op}} = a_\alpha S$, giving Eq. (38). The density need only be finite, Hermitian, and shape-compatible; idempotency, positivity, and electron count are optional diagnostics rather than hidden preconditions.

9.2 Covariant ambient projection

In the uniform covariant representation the canonical momentum and diamagnetic operator are

$$p_\alpha^{\text{can,cov}} = \Theta \odot (p_\alpha^0 + q a_\alpha S^0), \quad (69)$$

$$p_\alpha^{\text{dia,cov}} = -q a_\alpha S(t). \quad (70)$$

Their explicit a_α terms cancel, so the total ambient-projected mechanical operator reduces to

$$p_\alpha^{\text{mech,cov}} = \Theta \odot p_\alpha^0. \quad (71)$$

The resulting observable is named `ambient_projected_mechanical_current`. It is a useful common diagnostic, but it is not the P0+E1 variational source current in Eq. (61). The two must never share a generic stored `current` label.

10 Covariant energy and universal energy ledger

10.1 Covariant matter energy

The gauge-invariant P0/P0+E1 matter energy is evaluated from the inverse- dressed density in Eq. (44):

$$U_{\text{cov}}[P, t] = T_s[P^0] + E_{\text{eN}}[P^0] + E_{\text{H}}[P^0] + E_{\text{xc}}[P^0] + E_{\text{NN}}. \quad (72)$$

The E1 source-coupling expectation

$$U_{\text{E1,coupling}} = \text{Re Tr}(PV_{\text{E1}}) \quad (73)$$

is stored separately and is not included in Eq. (72). This is necessary for the Ward identity Eq. (64). Equivalent dressed-frame contractions are valuable independent validation identities, not alternative user-facing definitions.

10.2 Qualified totals

The energy observable records, when meaningful for a formulation:

1. canonical/paramagnetic kinetic, vector-potential-linear kinetic, diamagnetic kinetic, and their mechanical sum;
2. electron–nuclear, Hartree, combined exchange–correlation, and nuclear–nuclear energies;
3. electronic scalar EM coupling, fixed-nuclear scalar EM coupling, and their sum;
4. `energy_matter_total`;
5. `energy_generator_total`;
6. absorbed energy and accumulated source work;
7. analytic energy rate, source power, and Ward residuals.

The conventional DFT word *total* is retained only with a qualifier. There is no public `total_energy`, because Eqs. (33), (34), (42), and (64) show that different totals answer different physical questions during a pulse.

For every formulation,

$$\Delta U_{\text{abs}}(t) = U_{\text{matter}}(t) - U_{\text{matter}}(t_0) \quad (74)$$

is compared to independently accumulated source work. Fixed nuclei have no current and do not absorb power; their scalar coupling is nevertheless stored to establish origin cancellation in a neutral system.

10.3 Independent analytic energy derivative

For the pure-functional DFT energy, functional differentiation gives schematically

$$\frac{dU}{dt} = \text{Re Tr}[F_{\text{energy}}[P] \dot{P}] + \left. \frac{\partial U}{\partial t} \right|_P, \quad (75)$$

where $\dot{P}$ is obtained analytically from Eq. (14). This route is independent of the $\mathbf{E} \cdot \mathbf{J}$ contraction and therefore provides a non-tautological Ward test. Finite differences of sampled energies are only secondary convergence diagnostics.

At every accepted propagation interval, source work is advanced from the converged midpoint:

$$W_{n+1} = W_n + \Delta t \mathcal{P}_{\text{source}}(t_{n+1/2}). \quad (76)$$

The midpoint state and source already exist, so no additional Fock build is required. Work is accumulated every interval even when energy is not sampled there. A robust normalized Ward residual reports both absolute and relative scales and should decrease quadratically with Δt in the asymptotic regime.

Status — Planned/unvalidated. The draft tree has partial aggregate energy methods and ad hoc reconstruction of missing bare-VG kinetic terms. The complete qualified ledger and analytic functional energy derivative described here are not yet a reusable implementation.

11 Self-consistent exponential midpoint propagation

11.1 Why the connection matters numerically

For a fixed linear ODE, exponentiating a midpoint generator is routine. Here three facts interact:

1. the density determines the Kohn–Sham Hamiltonian;
2. the coordinate metric differs at the beginning and end of an interval;
3. the connection contains both an analytically integrable diagonal site piece and a residual projected piece that belongs in the dynamical generator.

An exponential that is unitary in one Euclidean or Löwdin-orthogonalized frame need not map the S_n norm to the S_{n+1} norm. Small defects can accumulate and feed back into the nonlinear density. The remedy is to respect parallel transport and cross-metric geometry explicitly, not to pretend that $S(t)$ is fixed.

11.2 Common nonlinear midpoint solve

Both production propagators use the same SCEM fixed point. Given an input midpoint density, build $(S_{1/2}, H_{\text{eom},1/2}, \omega_{1/2})$, construct a trial propagator, obtain an output midpoint density, and iterate to consistency. The convergence measure is

$$r_P = \frac{\|S_{1/2}^{1/2}(P_{\text{out}} - P_{\text{in}})S_{1/2}^{1/2}\|_F}{\max(N_e, 1)}. \quad (77)$$

The initial defaults are $r_P \leq 10^{-10}$, at most 50 iterations, and minimum damping 0.1. Start with a full Hermitian density update and reduce mixing when the residual worsens. Record the iteration count, final density residual, and an independent Hamiltonian residual. A failed midpoint is never accepted; tolerances and Δt are never silently changed.

11.3 Fixed-metric transport

For bare LG/VG, $S = S_0$ and $\omega = 0$. The midpoint generator is

$$\mathcal{A}_{1/2} = -\frac{i}{\hbar} S^{-1} H_{1/2}, \quad (78)$$

which is S -skew-Hermitian. Its rational map can therefore act directly in the nonorthogonal coordinates while preserving the S norm algebraically. A Löwdin transform is not required.

11.4 Connection-aware site-parallel frame

Write the analytically known diagonal site connection as $\sigma(t)$. Its exact parallel transport over one interval is

$$G_\sigma(t_{n+1}, t_n) = \exp\left[-\int_{t_n}^{t_{n+1}} \sigma(t) dt\right]. \quad (79)$$

For the diagonal Abelian uniform-field case this is elementwise and depends only on the known source. Pull the overlap, Hamiltonian, and residual connection into this site-parallel frame. At the self-consistent midpoint, the combined generator is

$$\bar{\mathcal{A}}_{1/2} = \bar{S}_{1/2}^{-1} \left[-\bar{\omega}_{\text{res},1/2} - \frac{i}{\hbar} \bar{H}_{1/2} \right]. \quad (80)$$

The E1 contribution in Eq. (57) is part of $\bar{\omega}_{\text{res}}$, not an auxiliary length-gauge Hamiltonian.

11.5 Rational exponential and metric correction

The production default is the Padé [2/2] map

$$R_{22}(Z) = \left(I - \frac{Z}{2} + \frac{Z^2}{12} \right)^{-1} \left(I + \frac{Z}{2} + \frac{Z^2}{12} \right), \quad Z = \Delta t \bar{\mathcal{A}}_{1/2}. \quad (81)$$

Cayley [1/1] is retained as a diagnostic. These are systematically improvable approximations to the frozen exponential[7], but the complete midpoint method remains globally second order unless the Hamiltonian/source quadrature and nonlinear construction are also raised in order.

After transforming the raw interval map back to AO coordinates, impose the cross-metric constraint by right Cholesky correction:

$$\begin{aligned} M &= U_{\text{raw}}^\dagger S_{n+1} U_{\text{raw}} = R_M^\dagger R_M, \\ S_n &= R_S^\dagger R_S, \\ Q &= R_M^{-1} R_S, \quad U = U_{\text{raw}} Q. \end{aligned} \quad (82)$$

Then $U^\dagger S_{n+1} U = S_n$ up to roundoff. Record the raw defect, corrected defect, and $\|Q - I\|$. This is a constraint correction to an already constructed physical transport; it is not propagation in an evolving symmetrically orthogonalized Löwdin basis.

Only roundoff-scale anti-Hermitian contamination may be symmetrized, under a separately recorded threshold. A larger defect is a hard failure.

12 Observable and storage model

12.1 Typed observable calculators

An observable declares:

- the formulation capabilities and operator/source dependencies it needs;
- whether it is evaluated at endpoints, midpoints, or events;
- its independent integer-step schedule;
- units, tensor shape, definition ID, and decomposition labels;
- whether evaluation requires a Fock build or may reuse midpoint data.

Unsupported observables fail during simulation construction. Energy is not computed every step by default. Energy, dipole/current, step diagnostics, checkpoints, and dense-matrix snapshots have independent schedules, while initial and final requested observables are always captured.

12.2 Canonical artifacts

The canonical numerical format is versioned HDF5:

Artifact	Purpose
<code>reference.h5</code>	Immutable backend-neutral prepared electronic reference
<code>trajectory.h5</code>	Canonical observable and diagnostic trajectory
<code>em_source.h5</code>	Optional full-resolution/per-provider source history
<code>checkpoint_<step>.h5</code>	Immutable restart state at an accepted boundary
<code>status.json</code>	Small atomic status record for detached monitoring

Top-level trajectory groups are

`/meta, /configuration, /reference, /time, /source,`
`/observables, /diagnostics, /events, /restart.`

Definition-specific observable paths are required; generic `/current`, `/dipole`, or `/energy` datasets are forbidden. Each stream owns step/time coordinates. Every dataset has unit and dimension attributes. Scalar streams are uncompressed; dense/checkpoint arrays are chunked on record boundaries with low-level gzip and shuffle.

Writers publish transactionally: write a `.partial` file, validate and flush it, set an internal completion marker, and atomically rename it. Completed artifacts are immutable. Controlled failures may publish `trajectory.failed.h5` with `complete=false`; a hard crash leaves only the partial artifact.

12.3 Restart and lineage

A checkpoint contains the dynamic state, occupations, global step/time, accumulated work, relevant nonlinear continuation data, and applied-event IDs. It contains no pickled PySCF or GPU object. Restart reconstructs the prepared reference from normalized input and verifies all fingerprints.

A resumed segment first reconstructs, writes, and authenticates the repeated boundary state n_r before advancing to $n_r + 1$. It records parent run ID, checkpoint hash, and global step offset. Analysis deduplicates the shared boundary only through this lineage. SIGINT/SIGTERM is handled at a safe accepted-step boundary; a half-converged midpoint is never checkpointed as a valid state.

12.4 Semantic identity and provenance

A deterministic simulation ID is the SHA-256 hash of losslessly normalized scientific configuration, array dtypes/shapes/bytes, source and reference fingerprints. Paths, timestamps, host names, and output schedules are excluded. A unique run ID distinguishes execution attempts. Full software, Git, dependency, CPU/GPU, and device provenance is stored. A requested GPU run never falls back to CPU.

13 Spectroscopy

Casida linear response and real-time kick spectra are postprocessing workflows. Casida roots, oscillator strengths, and transition directions are stored as structured numerical data[2]. A resonant comparison selects a polarization-bright root by an explicit threshold or root choice and recomputes it for every basis; otherwise an on-resonance/off-resonance comparison would be physically ambiguous.

For a kick κ , the canonical molecular response is

$$\alpha_{\alpha\beta}(\Omega) = \frac{\delta\tilde{\mu}_{\alpha}(\Omega)}{\kappa_{\beta}}. \quad (83)$$

The transform convention is

$$\tilde{f}(\Omega) = \int_0^T w(t)f(t)e^{+i\Omega t} dt. \quad (84)$$

The full complex result, sign, normalization, time origin, quadrature, window, damping, and zero padding are metadata, not hidden plotting choices. The default window is rectangular. Optional exponential damping $e^{-\eta t}$ stores η in energy units. The intrinsic frequency resolution is approximately $2\pi/T$; zero padding interpolates the sampled spectrum but does not improve this resolution. A current-domain transform supplies an independent frequency-domain check.

14 Reusable API and implementation boundaries

14.1 Static preparation and dynamic construction

The top-level Python workflow is intentionally small:

```
reference = prepare_reference(reference_config)
reference.save("reference.h5")

simulation = build_simulation(
    reference=reference,
    source=physical_source,
    formulation=formulation_config,
    propagation=propagation_config,
    observables=observable_config,
    output=output_config,
)
result = run(simulation)
resumed = resume(checkpoint_path, output=child_output)
trajectory = load_trajectory(path)
```

One prepared reference is reused by all formulations in a gauge comparison, preventing repeated SCF work and guaranteeing a common initial problem. References, configurations, operator bundles, sources, and formulation definitions are immutable. Backend caches and preallocated scratch arrays belong to a mutable `Workspace`; two simulations never share state or workspace.

14.2 Package domains

Package	Responsibility
<code>aion.config</code>	Immutable typed input, strict validation, normalization, units, and hashes.
<code>aion.backends</code>	Array protocol, CPU/GPU implementations, device residency, and workspaces.
<code>aion.electronic_structure</code>	PySCF RKS preparation, grids, reference reconstruction, core and derived operator bundles.
<code>aion.electromagnetism</code>	Physical sources, gauge representations, node/link samples, envelopes, compiled time series, and events.
<code>aion.formulations</code>	Bare LG/VG and P0/P0+E1 implementations of $(S, H_{\text{eom}}, \omega)$, formulation observables, and support checks.
<code>aion.propagation</code>	State protocols, common nonlinear SCEM, fixed-metric and connection-aware transport.
<code>aion.observables</code>	Typed dipole, current, mechanical-current, energy, power, continuity, and Ward calculators.
<code>aion.io</code>	HDF5 schemas, transactional writers, checkpoints, lineage, readers, and export.
<code>aion.workflows</code>	Preparation, simulation building, run/resume, comparisons, Casida, and kick spectra.

Dependencies are one-way: formulations consume references and sources; propagators consume the formulation protocol; observables inspect states and formulations; I/O receives records but never enters the physics. The CLI `aion prepare|run|resume|inspect|export` is a thin caller of the same typed API.

14.3 Prepared reference content

The immutable `reference.h5` contains normalized molecule/basis/ functional/grid input, converged coefficients and occupations, ground-state density, nuclear data, EM origin, exact quadrature-grid coordinates and weights, and the core operator bundle

$$\{S, T, V_{\text{nuc}}, \mathbf{r}, \mathbf{p}, \text{nuclear data}, f\}.$$

It does not store AO values on the grid and never pickles live objects. Loading reconstructs PySCF/backend models and verifies fingerprints. Atomic projectors/topology and P0/E1 tensors are independently versioned derived bundles with their own invalidation boundaries.

15 Present draft versus the 0.2 target

Status — Current-tree assessment after WP0. The research draft was archived and verified on 8 September 2026. Its molecule-specific campaigns, runners, postprocessing, historical notes, and large results were moved with checksum manifests to the separate CALCULATIONS repository. The reusable draft modules listed below remain in Aion as individually reviewable source material; they are not an API that 0.2 must preserve.

The principal reusable draft components are:

Current module	Implemented research functionality
<code>src/aion/cn_tddft.py</code>	Bare length-gauge propagation, LG kick, density/dipole utilities, and fixed-metric SCEM; it still contains older propagation paths and a coarse energy interface.
<code>src/aion/velocity_gauge.py</code>	Bare local velocity-gauge Hamiltonian, reduced-vector-potential pulse, discontinuous VG kick, and mechanical current.
<code>src/aion/pyscf_p0.py</code>	CPU/GPU inverse density dressing, PySCF potential/energy construction, and forward Hamiltonian dressing; live mean-field state is not yet separated into a backend-neutral reference artifact.
<code>src/aion/p0_e1.py</code>	Central dipoles, E1 variational potential, connection contribution, covariant dipole, and analytic E1-current terms.
<code>src/aion/observables.py</code>	P0 site charge, analytic charge derivative, pair/graph current, continuity, and graph power.
<code>src/aion/connection_cayley.py</code>	Connection-aware SCEM experiments, Cayley/Padé transport, and Cholesky metric correction.
<code>src/aion/p0_diagnostics.py</code>	Flat research diagnostic rows, including partial current/power/energy quantities.

Important missing reusable layers are:

- a unified immutable reference and source/simulation builder;
- formulation-owned typed primary and diagnostic observables;
- the generic mechanical-current operator API;
- the complete qualified energy ledger and independent analytic dU/dt ;
- a single shared SCEM nonlinear engine behind both transports;

- versioned transactional HDF5 with exact restart lineage;
- a thin common CLI and reusable spectroscopy outputs.

At present, some comparison scripts reconstruct missing VG linear and diamagnetic kinetic energies, calculate bare-LG momentum current, or assemble covariant mechanical diagnostics after propagation. Plotting scripts also encode observable definitions. In 0.2 these calculations belong to `aion.observables`; campaigns select and plot stored definitions but do not rederive them.

16 Validation requirements

16.1 Hierarchy

The validation fixtures have distinct purposes:

Synthetic matrices

Exact signs, Hermiticity, metric compatibility, graph incidence/orientation, gauge covariance, rational-map algebra, and cross-metric transport.

H₂ Zero field, molecular symmetry, bare-gauge smoke tests, exact kicks, and compact timestep convergence.

LiH Nonzero E1 response, gauge/origin transformations, variational current, continuity, Ward energy, connection transport, restart, and CPU/GPU parity.

NH₃ Integrated release validation only: aug-cc-pVTZ, five driven cycles plus five field-free cycles, all four formulations, accepted step and a shorter half-step interval.

CO, ammonia/PNA production studies, and basis-convergence campaigns are external scientific calculations, not unit-test decisions. Existing draft trajectories are evidence, not golden truth.

16.2 Independent internal checks

The principal assertions are:

1. $P = P^\dagger$, electron number and occupied-state metric normalization;
2. Eq. (13) and cross-metric preservation;
3. explicit gauge-function transformations and EMF invariance;
4. translation of the EM origin, including electronic/nuclear cancellation for neutral systems;
5. Eq. (62) on every node and consistency of pair signs with Eq. (21);
6. directional derivatives of the variational source action versus Eq. (61);
7. analytic Eq. (63), independent of finite differences;
8. energy-component sums and analytic Eq. (75) versus source power;
9. approximately second-order reduction of trajectory and Ward errors under step halving;
10. uninterrupted versus restarted trajectory identity;

11. CPU/GPU parity for state, observables, energy components, invariants, and backend residency.

Full-array pinned trajectories are not the primary oracle. Selected numerical endpoints are retained only when they provide an independent protection.

16.3 GPU and performance policy

Every milestone that changes executable numerical physics must run its applicable tests on the physical GPU. The execution sandbox may hide the device; this is handled by requesting elevated execution, never by omitting GPU tests or substituting CPU-only tests. A requested GPU backend fails if unavailable.

The fast test suite uses up to eight workers with one BLAS thread each. Molecular CPU tests run serially with up to eight BLAS threads, and GPU tests run serially. Preparation, Fock construction, steady step time, memory, I/O, and host transfers are measured separately. An unexplained steady-step regression of roughly ten percent or more is flagged, but correctness and physics gates take priority.

17 Conclusions

The bare and covariant formulations solve related physical problems but do not share interchangeable current or energy definitions. Bare LG uses the analytic dipole derivative as its primary current; bare VG uses paramagnetic plus diamagnetic mechanical current and requires the complete mechanical kinetic energy; P0+E1 uses the variational source current implied by its projected connection. A common ambient-projected mechanical current is an important diagnostic, not a replacement for the covariant source current.

The difficult numerical feature of P0+E1 is not ignorance of the connection: the connection is analytically known from prescribed fields and static matrix elements. The issue is faithful finite-step transport between distinct metrics while the nonlinear Hamiltonian is determined at the midpoint. Exact site parallel transport, a combined residual generator, Padé midpoint propagation, and a right Cholesky cross-metric correction address that geometry without an evolving Löwdin frame.

The planned 0.2 architecture makes these physics definitions reusable and auditable. It separates immutable reference preparation, physical sources, formulations, propagation, observables, and storage; it records qualified energy/current definitions; and it treats conservation, Ward identities, convergence, restart, and CPU/GPU parity as release gates. The preservation and repository-separation gate WP0 is complete. Numerical and API implementation still requires separate authorization.

Rebuilding this document

From the `aion/docs` directory, the PDF is rebuilt and all references are resolved with the single shell command

```
for pass in 1 2 3; do \  
  /usr/bin/pdflatex -interaction=nonstopmode \  
  -halt-on-error theory_and_implementation.tex; \  
done
```

The generated auxiliary, log, outline, and table-of-contents files are build artifacts; the tracked deliverables are the TeX source and PDF.

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